What is there in an Indian Thali? - Supplementary

A Weight-Regression Network Variants

We evaluate three key design choices in our weight–regression pipeline: (i) **attention placement and type**, (ii) **geometric ROI level scalars**, and (iii) **RGB+Depth feature fusion**. All models share the same training protocol, optimizer, and data splits; only the specified component is varied via the ablation flags in our training script. Results are consolidated in Table 1.

A.0.1 Attention Ablation: CBAM (pre-ROI) vs. SelfAttention (post-ROI). **Motivation.** Attention can improve weight estimation by refining features to emphasize salient channels and spatial regions. We test two *isolated* attention variants, each applied at a different point in the pipeline.

What varies.

- *None (Baseline)*: ResNet-50 + MLP head, no attention module.
- *CBAM only*: Convolutional Block Attention Module applied **before RoI Align** on encoder features; no self-attention block used.
- *Self only*: Self-attention block applied **after RoI Align** on ROI-aligned features (post-fusion); no CBAM used.

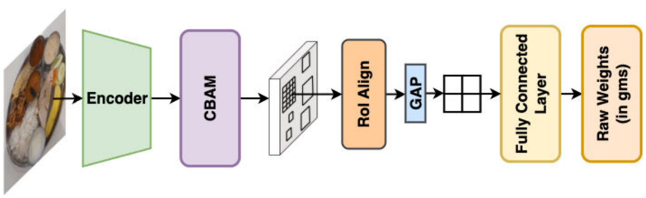


Figure 1: Architecture of the *CBAM only* variant. The CBAM block is applied to the encoder output (pre-ROI) without a self-attention block.

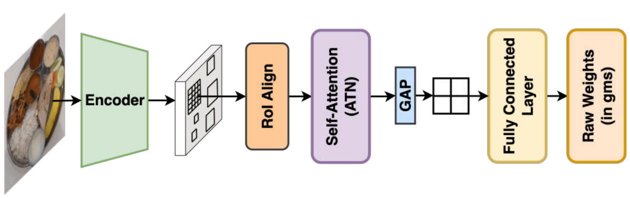


Figure 2: Architecture of the *Self only* variant. The self-attention block operates on ROI-aligned features (post-ROI) without a CBAM block.

Results (from Table 1).

- *CBAM only* substantially degrades performance (e.g., Calories MAE **20.439** vs. 16.621 baseline; Weight MAE **18.557** vs. 14.514 baseline).
- *Self only* closely matches the baseline (Calories MAE 16.855; Weight MAE 14.771) with only minor increases in %Err.

Observations.

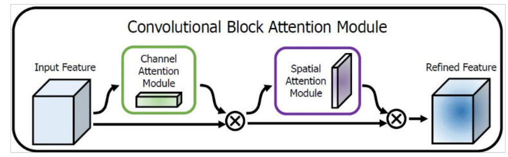
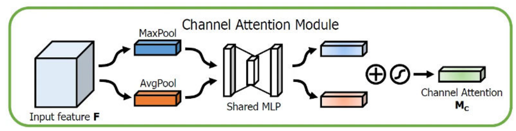
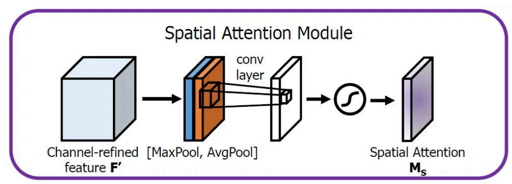


Figure 3: Reference CBAM architecture, consisting of sequential channel and spatial attention stages.



(a) Channel attention: global average & max pooling → shared MLP → channel weights.



(b) Spatial attention: pooled descriptors → convolution → spatial attention map.

Figure 4: CBAM internal components used in the *CBAM only* variant.

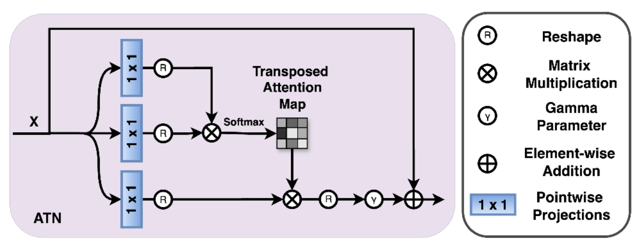


Figure 5: Architecture of the self-attention block in the *Self only* variant. Q, K, and V projections are computed from the ROI-aligned feature map X; attention weights are obtained from $QK^{\top}$, applied to V, scaled by a learnable γ, and added back to X via a residual connection.

- Applying attention **post-ROI** (Self) preserves ROI feature integrity and avoids suppressing useful context, unlike pre-ROI CBAM.
- Stacking CBAM+Self is unlikely to help, as isolated Self already preserves baseline performance without degradation.

A.0.2 Geometric ROI-Level Scalars. **Motivation.** Beyond visual features, geometric cues can provide stable size and volumetric

Nutrient	Metric	None (Baseline)	Area Only	Both	DepthMod	CBAM	Self
Calories	MAE	16.621	16.573	16.482	16.728	20.439	16.855
	MSE	803.953	807.499	803.495	822.426	982.039	819.345
	% Error	15.764	15.718	15.632	15.865	19.384	15.985
Carbs	MAE (g)	2.764	2.752	2.738	2.782	3.353	2.783
	MSE	26.294	26.126	26.148	26.757	31.306	26.123
	% Error	16.597	16.524	16.438	16.707	20.135	16.713
Protein	MAE (g)	0.607	0.603	0.598	0.611	0.746	0.615
	MSE	1.215	1.206	1.174	1.220	1.506	1.210
	% Error	15.654	15.535	15.418	15.735	19.232	15.850
Fat	MAE (g)	0.400	0.401	0.399	0.402	0.511	0.415
	MSE	0.581	0.618	0.604	0.618	0.771	0.658
	% Error	13.746	13.788	13.710	13.824	17.592	14.258
Weight	MAE (g)	14.514	14.439	14.529	14.598	18.557	14.771
	MSE	501.400	500.668	507.438	511.054	655.352	520.232
	% Error	14.624	14.549	14.640	14.709	18.698	14.883

Table 1: Ablation: Weight and nutrient regression network variants.

context for weight prediction. We test two scalars: *Normalized Area* and *Mean Depth*.

Description. Given an ROI mask $m \in \{0,1\}^{H\times W}$ and its normalized depth map $D \in [0,1]^{H\times W}$:

$$\text{Normalized Area} = \frac{\sum_{u,v} m_{u,v}}{H \times W}, \tag{1}$$

$$\text{Mean Depth} = \frac{\sum_{u,v} D_{u,v} \cdot m_{u,v}}{\sum_{u,v} m_{u,v} + \varepsilon}. \tag{2}$$

Normalized Area is resolution-invariant and reflects portion size; *Mean Depth* captures average height/volume.

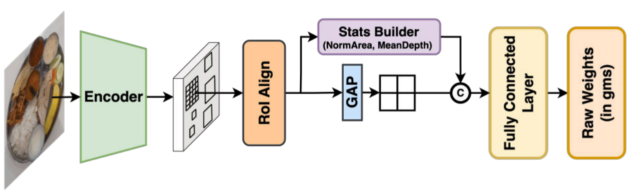


Figure 6: Integration of geometric scalars into the weight-regression network. Pooled ROI features are concatenated with *Normalized Area* and optionally *Mean Depth* before the MLP regression head.

Results (from Table 1).

- *Area only* yields slight improvements over baseline (Calories MAE 16.573 vs. 16.621; Weight MAE 14.439 vs. 14.514).
- *Both* (Area + Mean Depth) achieves the best results (Calories MAE **16.482**; %Err **15.632**) with consistent nutrient-level gains.

Observations.

- *Normalized Area* offers a stable cue that is independent of resolution.
- *Mean Depth* adds volumetric context, complementing area.
- Combining both yields modest but consistent improvements without altering the main feature extraction process.

A.0.3 Depth Fusion (RGB + Depth). **Motivation.** Depth cues can provide complementary 3D structure beyond RGB appearance. We evaluate late fusion of monocular depth with RGB features.

Description. Depth maps are predicted with Depth Anything V2 (Small), normalized to $[0, 1]$, and treated as single–channel features. RoI Align is applied separately to RGB encoder features and to the depth map; the ROI tensors are concatenated along the channel dimension before global average pooling and the MLP head. No attention modules are used in this variant.

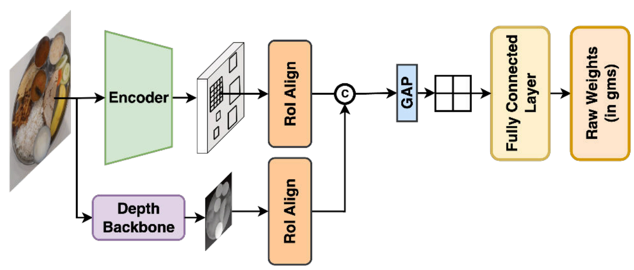


Figure 7: Architecture of the depth fusion variant. ROI-aligned RGB and single–channel depth features are concatenated along the channel dimension before pooling and regression.

Results (from Table 1).

- *DepthMod* shows minimal change from baseline (Calories MAE 16.728 vs. 16.621; Weight MAE 14.598 vs. 14.514; Calories %Err 15.865 vs. 15.764).

Observations.

- Late fusion of single–channel depth without a dedicated encoder yields marginal benefits, likely due to depth noise or weak correlation to plated mass at ROI scale.
- Future improvements may come from learned cross-modal fusion or shallow depth encoders.

B Future Work

Future research will focus on advancing the scalability and generalization of our approach. We aim to:

- Automate prototype creation and updating, reducing reliance on manual labeling.
- Enhance robustness to visual complexity, enabling reliable segmentation in more challenging real-world conditions.

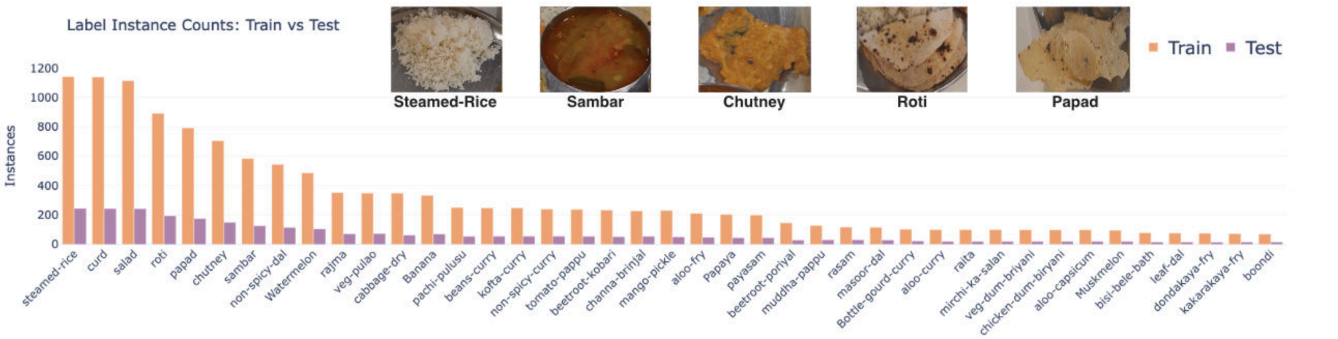


Figure 8: Distribution of labeled dish instances in the Weight Estimation Dataset (WED), separated into training and testing splits. The dataset exhibits a long-tail pattern, with a few dominant classes (e.g., steamed rice, sambar) and many underrepresented items, reflecting real-world cafeteria serving frequency.

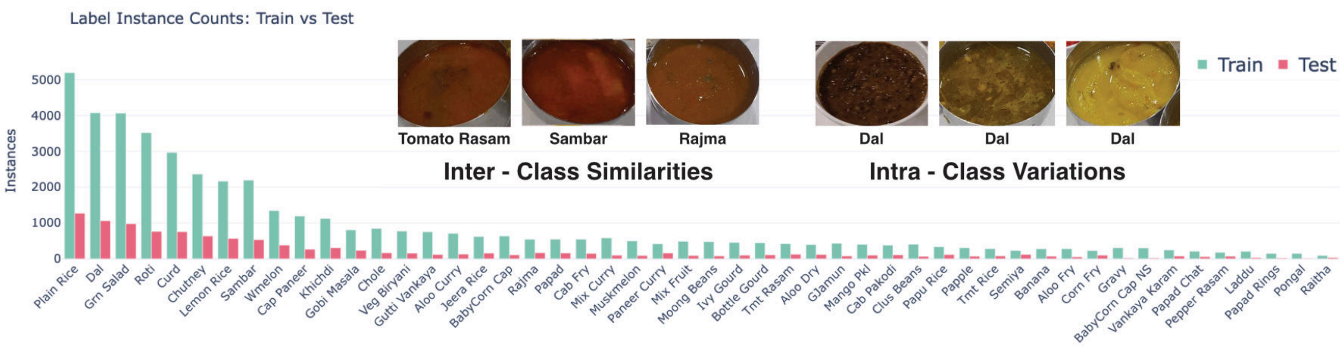


Figure 9: Label instance counts for the Indian *Thali* dataset, comparing the distribution of classes in the training (teal) and testing (pink) sets. Classes are sorted in descending frequency, showing a pronounced long-tail distribution where a few categories, while many others have relatively few examples. The figure also highlights visual challenges in the dataset: inter-class similarities (e.g., visually similar items like *Tomato Rasam*, *Sambar*, and *Rajma*) and intra-class variations (e.g., substantial appearance differences within the Dal class). These characteristics underscore the difficulty of achieving robust recognition performance in this domain.

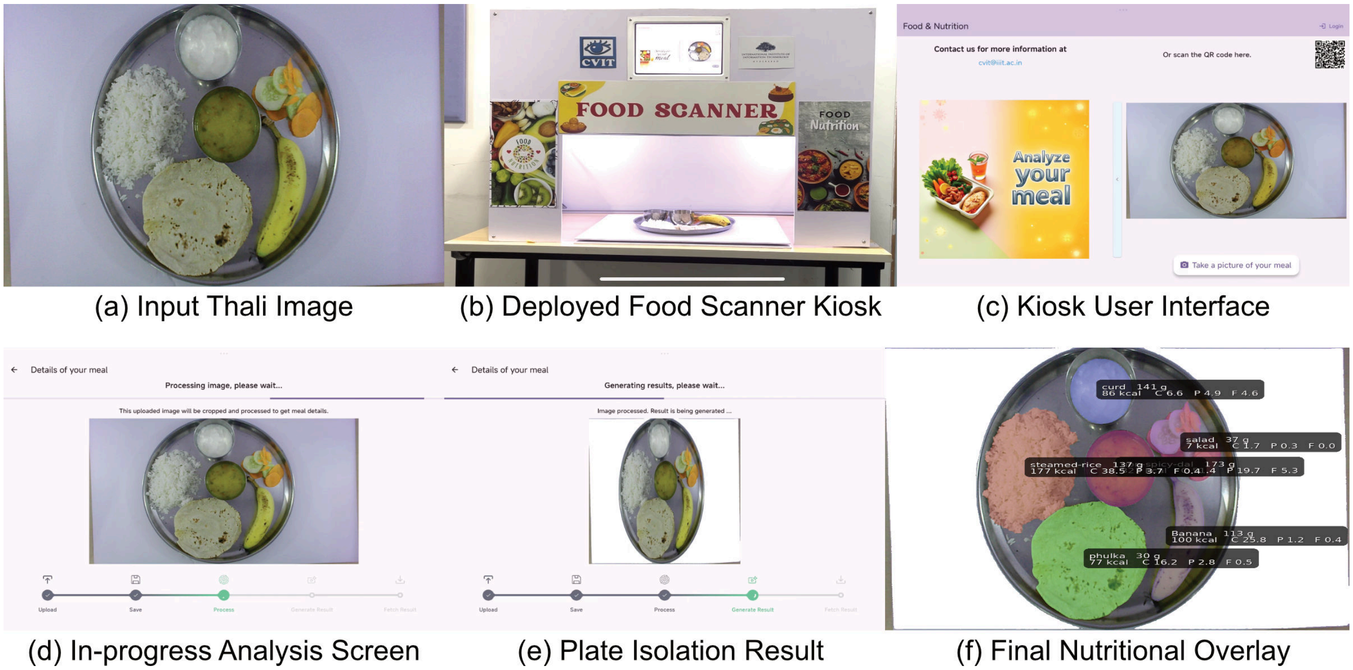


Figure 10: The complete user workflow and interface of the deployed Food Scanner kiosk. A user places their thali (a) at the station (b) and begins the process via the tablet's welcome screen (c). The interface provides feedback during the analysis (d) and shows intermediate technical steps, such as background removal (e). The final output presents a complete nutritional breakdown, with segmentation masks and per-item data for every component on the plate (f).

Figure 11: Granular, per-item analysis of a complete thali. The figure demonstrates the final output, providing a detailed breakdown for each individual food item on the plate. Each row displays the segmentation mask on the full thali (left) and a corresponding zoomed-in view with the specific nutritional data (right).

#	Class	FCN	DeeplabV3+ (R50)	SegNeXt (L)	Mask2Former (R50)	SegFormer (B3)	Mask2Former (SwinT)	ClipSeg (ZS)	ClipSeg (FT)	Ours
1	Aloo Dry fry	62.01	76.62	94.97	95.44	94.89	95.45	22.39	91.48	78.82
2	Avakaya Muddha Papu Rice	33.06	91.29	96.12	96.50	95.70	96.50	5.40	93.69	58.92
3	Baby-Corn & Capsicum-Dry	38.30	74.01	92.66	93.37	92.17	93.41	25.70	89.20	81.03
4	Cabbage Pakodi	25.22	80.82	88.84	89.47	88.73	89.59	33.17	85.85	87.63
5	Cabbage fry	34.91	52.61	94.13	92.96	94.05	94.73	30.31	90.04	90.75
6	Capsicum Paneer Curry	59.29	60.06	92.55	92.60	92.11	92.90	0.34	90.96	52.09
7	Chakar-Pongal	31.14	37.75	78.16	78.00	77.66	79.00	13.27	73.99	77.76
8	Chole-Masala	85.74	87.05	93.36	93.74	92.60	94.27	34.05	92.18	68.55
9	Cluster Beans Curry	28.83	34.85	90.47	88.40	90.53	90.92	19.51	86.93	78.04
10	Cucumber-Raitha	74.30	73.11	79.67	77.87	78.50	78.48	3.59	82.34	59.53
11	Gobi Masala Curry	39.73	75.62	93.23	93.79	92.10	93.71	10.09	90.97	60.17
12	Gutti Vankaya Curry	89.85	91.41	93.00	93.45	92.79	93.76	21.02	90.86	63.47
13	Jeera Rice	2.54	0.55	94.54	94.74	94.32	95.09	28.96	92.60	88.62
14	Mixed Curry	73.14	86.46	93.89	94.37	93.71	94.15	1.86	91.02	85.98
15	Muskmelon	22.88	83.26	94.38	94.96	93.66	95.02	45.22	90.02	90.58
16	Rajma	41.81	82.98	90.93	91.15	89.95	91.99	12.83	90.88	82.38
17	Rasgulla	47.33	79.02	96.21	96.43	94.29	96.83	10.23	92.18	88.15
18	Sambar	87.07	87.51	93.67	93.95	93.58	94.54	19.12	92.06	81.11
19	Tomato Rasam	71.79	88.00	94.32	94.86	93.51	94.60	33.59	92.56	83.19
20	Vankaya-Ali-Karam	76.61	62.30	89.16	89.77	88.09	90.02	0.00	86.55	61.45
21	Veg-Biriyani	42.99	42.99	92.99	93.51	92.83	93.45	27.61	92.35	89.57
22	aloo-curry	16.43	74.44	94.58	95.18	93.91	95.25	0.02	93.08	68.59
23	curd	84.83	92.94	94.18	94.75	93.76	94.81	7.53	92.17	81.49
24	dal	87.20	86.55	93.71	93.96	93.11	94.18	8.12	92.00	72.29
25	fresh-chutney	59.92	78.91	91.57	92.32	91.30	92.28	7.31	85.63	75.04
26	green-salad	91.33	91.53	93.31	93.85	92.86	93.94	10.02	87.96	85.01
27	Moong-Beans-Curry	0.93	90.82	94.53	95.17	94.11	95.06	13.36	91.25	93.56
28	khichdi	70.00	91.20	94.50	94.73	94.30	95.01	14.42	91.86	82.83
29	lemon-rice	82.67	76.78	94.57	94.97	94.69	95.05	31.10	92.87	93.11
30	live-roti-with-ghee	90.27	96.94	97.73	98.11	97.59	98.09	57.07	95.69	93.53
31	non-spicy-curry-bottle-gourd	7.97	11.35	94.01	94.67	93.53	94.64	0.83	90.33	92.64
32	papad	16.98	85.50	93.63	93.71	93.52	94.01	9.83	88.76	65.71
33	plain-rice	94.33	94.44	95.28	95.54	95.19	95.74	68.62	93.69	93.67
34	watermelon	64.14	91.31	95.10	95.80	94.86	95.68	48.66	90.88	83.69
35	Aloo-Fry	30.65	84.14	91.82	92.74	91.47	92.73	4.56	86.71	37.25
36	Banana	8.52	88.46	96.31	97.13	94.88	96.30	60.65	90.72	93.24
37	Mix-Fruit	16.89	93.15	95.52	96.10	94.95	96.19	20.35	90.72	88.94
38	Non-Spicy-Baby-Corn & Capsicum-Dry	76.80	42.64	88.57	89.66	89.02	90.12	0.58	84.86	85.33
39	Sweet	72.55	66.39	90.53	93.62	90.84	90.23	0.00	86.34	68.12
40	Tomato-Rice	14.06	45.40	95.76	96.34	95.54	96.36	1.27	93.67	89.73
41	fried-papad-rings	88.60	89.27	91.73	90.90	90.37	90.69	40.52	83.52	85.81
42	gravy	18.13	74.05	91.25	91.91	91.14	91.93	4.87	90.75	59.29
43	ivy-gourd-fry	73.45	88.49	94.41	95.57	95.06	95.59	4.95	91.20	89.93
44	mango-pickle	59.75	70.37	91.01	92.05	91.26	92.05	1.51	85.15	60.90
45	papad-chat	84.73	86.02	92.05	93.02	91.83	93.02	1.66	89.17	79.74
46	pepper-rasam	50.06	37.42	95.31	95.32	94.27	95.35	19.76	92.94	68.89
47	pineapple	47.63	89.99	94.04	95.09	94.14	95.12	25.44	89.33	77.92
48	corn-fry	69.85	60.16	88.19	89.78	88.78	89.74	2.74	85.78	66.89
49	paneer-curry	89.11	80.46	91.64	92.38	90.78	93.28	0.25	91.84	87.39
50	semiya	90.07	89.69	91.84	92.02	91.95	92.18	15.69	90.13	88.45
	Mean	54.53	73.94	92.68	93.11	92.30	93.26	17.60	89.75	78.34

Table 2: Final IoU (%) Comparison of All Segmentation Models Across Food Classes.

Figure 12: Custom-built edge-deployed Food Scanner station used for real-time *thali* analysis in our lab. The system integrates controlled lighting and an overhead camera to capture plate images under consistent conditions for zero-shot segmentation, prototype-based classification, and weight estimation.

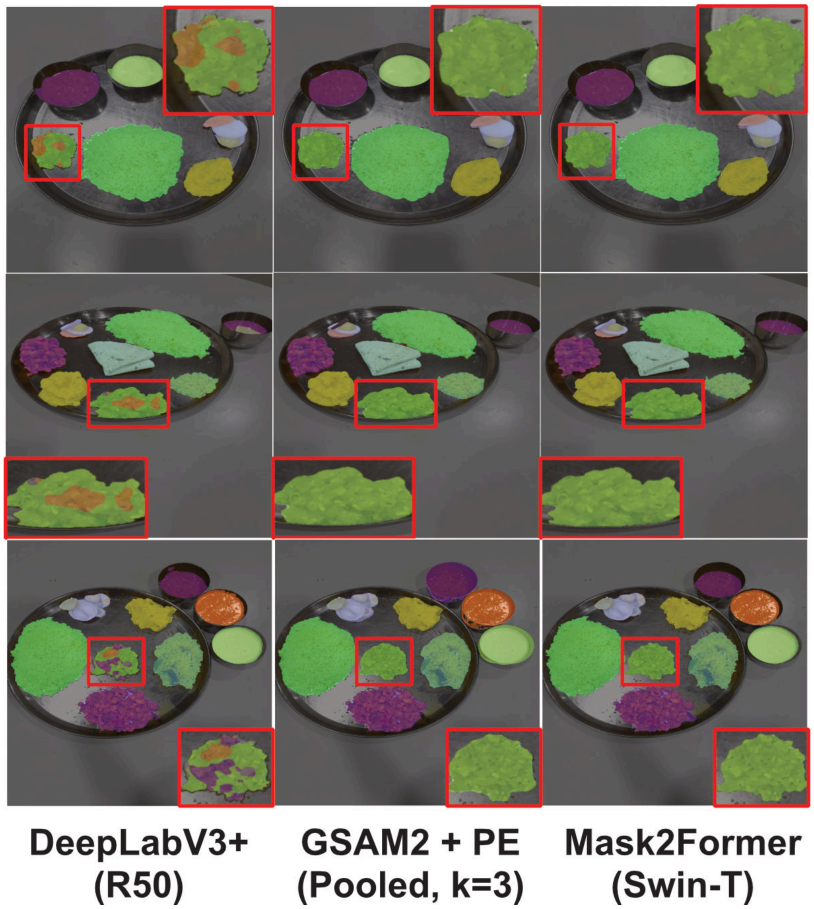


Figure 13: Qualitative segmentation results on the Indian *Thali* dataset. Our training-free *GSAM2 + PE* pipeline (middle column) achieves segmentation quality comparable to fully-trained baselines *DeepLabV3+ (R50)* (left) and *Mask2Former (Swin-T)* (right), while preserving fine object boundaries and reducing misclassification in ambiguous regions.